Structural Certificates for Prediction Depth

Frozen and transported control in one-sided cellular automata

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Abstract. We classify the prediction depth of one-sided cellular automata whose frozen control words gate a stay/hop data update over $\mathbb{F}_2$. The input is the explicit initial cone and the output is the data bit at its apex. For each fixed rule, a single logspace-uniform family of polynomial-size De Morgan circuits realizes the classified depth upper bound. The variable-guard regime has depth $\Theta(\log(T+1))$, and homogeneous Pascal prediction has depth within a factor of three of the number of ones in the binary expansion of T. We also analyze a moving-control model in which a single data path accumulates Boolean guards. At positive times, equal control and data velocities give depth one, whereas unequal velocities give exact depth $\lceil \log_2(T+1) \rceil$. The finite statements follow from Lean 4 proofs of the dynamics, circuit correctness, and the fan-in lower bounds.

Keywords. Cellular automata, circuit depth, logspace uniformity, Lean 4.

1 Introduction

Algebraic structure can make a cellular automaton's future state accessible to much shallower circuits than layer-by-layer simulation [1]. We ask which fixed guards in a one-sided, two-track automaton admit constant-depth prediction and how depth depends on the observation time.

For frozen controls, path counting gives a uniform logarithmic-depth construction and input restrictions give lower bounds. The resulting four regimes include a homogeneous rule whose depth can drop between consecutive times (Theorem 2). For transported controls, relative velocity determines the number of distinct initial controls encountered along the data path. A balanced conjunction attains the resulting lower bound (Theorem 3). Both classifications are formalized in Lean 4.

2 Preliminaries

A local rule on configurations over $\mathbb{Z}$ is iterated T times. We ask for one data bit, at site zero. Each model in the main results has a one-sided radius-one update, so its initial window consists of sites $0, \ldots, T$. For the frozen model the input contains $q(T+1)$ control bits followed by $T+1$ data bits. The control width q and the two guard circuits are fixed independently of T. The legal word lengths are $n = (q+1)(T+1)$ and determine T uniquely. The language consists of the legal words whose predicted bit is one, with all other lengths rejected.

Our upper bounds use Boolean AND and OR gates of fan-in two, NOT gates, and constants. An input wire has depth zero. Gates can share outputs and connect arbitrary sites. The per-time lower bounds apply to any Boolean gates of fan-in at most two in this wiring model. For a fixed rule, write $D(T)$ for the minimum De Morgan circuit depth on the legal window at time T and write $\ell(m) = \lceil \log_2 m \rceil$ for $m \geq 1$. Uniform circuit families are indexed by input length. A deterministic logspace transducer receives the length in unary and writes the complete circuit code, including its gates, wiring, and output. We use logspace uniformity throughout the statements about NC.

A coordinate is *essential* if two assignments that differ only at that coordinate give different outputs.

Essential-input bound. A depth-d circuit of fan-in at most b has at most b^d essential coordinates. For $b \geq 2$, a function with $m \geq 1$ essential coordinates therefore requires depth at least $\lceil \log_b m \rceil$.

Proof. Only input wires in the backwards cone of the output can affect it. At depth zero the cone has at most one input. At the next depth a gate has at most b predecessor cones, so induction bounds the number of input wires by b^d. In a shared circuit, counting a repeated predecessor more than once only increases the bound. □

3 Frozen-control automata

3.1 The iterate formula

Let $a_i \in \{0,1\}^q$ be a control word, $x_i \in \mathbb{F}_2$ an initial data bit, and $s, h : \{0,1\}^q \to \{0,1\}$ the stay and hop guards. The synchronous update is

$$a_i' = a_i, \quad x_i' = s(a_i)x_i + h(a_i)x_{i+1}, \tag{1}$$

where the data arithmetic is in $\mathbb{F}_2$. A backwards path ending at site k has k hops and $T-k$ stays. Put

$$e_k = \prod_{j=0}^{k-1} h(a_j), \quad w_k = \sum_{j=0}^{k} s(a_j), \tag{2}$$

where the second sum is in the natural numbers. For natural w, r, write $M(w,r)$ for the number of weak compositions of r into w parts. For $w > 0$, it is $\binom{w+r-1}{r}$, and $M(0,r)$ is one at $r = 0$ and zero otherwise.

Theorem 1 (Iterate formula). *For every initial configuration and time T, the data bit at site zero is*

$$x_0(T) = \sum_{k=0}^{T} e_k (M(w_k, T-k) \bmod 2) x_k. \tag{3}$$

The same identity holds at any site after translating the indices.

Proof. Expanding the local recurrence once per step expresses the output as a sum over backwards paths. Each hop must be enabled, contributing the factor e_k. A path visits the sites 0 through k in order. The number of stays allocated to each enabled site determines the path, including possible stays after the last hop. There are $M(w_k, T-k)$ such allocations. Their contributions add in $\mathbb{F}_2$, giving Equation 3. The Lean proof establishes the corresponding coefficient recurrence by induction on the iterate of Equation 1. The formula involves only the initial window. □

3.2 Depth classification

Write $p(T)$ for the number of ones in the binary expansion of T, and d_s, d_h for the depths of the fixed guard circuits. A guard varies if it takes both Boolean values. All the bounds below are finite, including $T = 0$.

Guard condition	Lower bound	Uniform upper bound
$h \equiv 0$	0	$1 + d_s$
$h \equiv 1$, $s \equiv 0$	0	0
h varies, or $h \equiv 1$ and s varies	$\ell(T+1)$	$104(\ell(T+3)+1) + \max(d_s, d_h)$
$s \equiv h \equiv 1$	$p(T)$	$3p(T)$

Table 1: Depth bounds at legal lengths for fixed frozen guards.

Theorem 2 (Uniform classification). *For each fixed pair of guard circuits there is one logspace-uniform, polynomial-size De Morgan family that decides the whole language and achieves the appropriate upper bound in Table 1 at every legal length. It has depth $O(\log(n+2))$ at all lengths. The lower column bounds $D(T)$ and holds even for arbitrary fan-in-two Boolean gates. The language is in logspace-uniform NC^0 exactly when $h \equiv 0$, or when $h \equiv 1$ and $s \equiv 0$.*

3.3 Proof of the uniform classification

Local and shift cases. For $h \equiv 0$ and $T > 0$, the answer is $s(a_0)x_0$ by idempotence. The pure shift case returns x_T. At $T = 0$ every rule returns x_0.

Varying guards. When h varies, set all data bits except x_T to zero. The surviving term in Equation 3 is $x_T \prod_{j=0}^{T-1} h(a_j)$, regardless of s. A nonconstant Boolean guard has an essential bit, as seen along a single-bit path between two words with different guard values. Each of the first T control words contributes one essential bit on this restriction, and x_T contributes one more. When $h \equiv 1$ and s varies, fix $s(a_0) = 1$ and $s(a_i) = 0$ at sites 1 through T. Then $w_k = 1$ for every k and Equation 3 is parity of all $T + 1$ data bits. Both cases give the same lower bound by essential coordinates.

For the upper bound, evaluate the guards in parallel and compute the prefix weights w_k and products e_k from Theorem 1. The weight circuit uses balanced carry-save reduction followed by binary addition. For $w_k > 0$, the parity of $M(w_k, T-k)$ tests whether every set bit of $T-k$ is set in $w_k + T - k - 1$. At $w_k = 0$ use the zero-part convention. The uniform construction uses a logspace-generated table of these coefficients, selected by the computed prefix weights. A balanced parity reduction combines the data contributions. The depth bound in Table 1 accounts for the guard, arithmetic, selection, and reduction stages.

Homogeneous Pascal rule. For $s \equiv h \equiv 1$, the coefficients are $\binom{T}{k} \bmod 2$. Over $\mathbb{F}_2$, the identity

$$(1+z)^T = \prod_{j:\mathrm{bit}_{j(T)}=1} \left(1 + z^{2^j}\right) \tag{4}$$

shows that precisely the binary submasks of T occur. Their $2^{p(T)}$ data bits are essential. The logarithmic lower bound is therefore $p(T)$. A balanced XOR tree reduces those leaves in $p(T)$ rounds. Each XOR can be implemented with at most three De Morgan layers. The uniform generator scans $k = 0, \ldots, T$, emits the submask wires, and serializes a regular XOR reducer with $5p(T)2^{p(T)}$ gates.

A single uniform family. The guard circuits and the selected branch are fixed. The logspace transducer writes that branch's predictor at legal lengths and a rejecting circuit at other lengths. Its polynomial-length output bounds the number of gates. The local and shift branches have a constant bound. In the varying branch the lower bound grows with T. For the homogeneous branch it grows along the times $T = 2^r - 1$. This proves the constant-depth characterization. □

At $T = 2^r - 1$ the homogeneous rule needs at least r layers, while at the next time $T = 2^r$ it has a circuit of depth at most three. For $r > 3$ this gives a strict drop in minimum depth. The accompanying classifier tests constancy of the guards by traversing control words by prefix. A constancy test stops at its first counterexample and can visit 2^q words in the worst case.

4 Transported controls

We now give the control track its own dynamics. Fix velocities $u, v \in \{0, 1\}$ and a Boolean control and data track (c_i, d_i). Define the synchronous rule

$$c_i' = c_{i+u}, \quad d_i' = c_i \wedge d_{i+v}. \tag{5}$$

The window has $2(T+1)$ bits. With a single backwards data path, the data at site i and time T equals

$$d_{i+vT} \wedge \bigwedge_{k=0}^{T-1} c_{i+kv+(T-1-k)u}, \tag{6}$$

where the right-hand side uses initial bits and the conjunction is empty at $T = 0$.

Theorem 3 (Exact depth under transport). *Depth is zero at $T = 0$. For $T > 0$, it is exactly one if $u = v$ and exactly $\ell(T+1)$ if $u \neq v$. The formal upper-bound construction uses a padded AND tree with at most $2m$ binary AND gates for m essential inputs. The lower bound applies to all fan-in-two Boolean circuits predicting the data bit.*

Proof. At the k-th step of the backwards path we read control at site $i + kv$ and time $T - 1 - k$. The initial site of that control bit is $i + kv + (T - 1 - k)u$, which proves Equation 6. At $T = 0$ the answer is the input wire d_i. At positive times with equal velocities all control indices coincide, leaving a conjunction of one control bit and one data bit. For unequal velocities the initial control indices are $i, ..., i + T - 1$ in forward or reverse order. Every distinct factor is essential, since flipping it with all other factors set to one changes the output. Thus there are two essential coordinates in the equal case and $T + 1$ in the unequal case. For m factors, a balanced AND tree of height $\ell(m)$ attains the essential-input lower bound. Padding to the next power of two uses $2^{\ell(m)} - 1 \leq 2m$ binary gates. □

The pair $(u, v) = (0, 0)$ is the frozen no-hop case with stay guard c_i. Changing only u to one gives logarithmic depth although the data never hops. For $(u, v) = (1, 1)$ the data and control both move, yet their equal velocity gives constant depth. For this single-path rule, the relevant count is the number of distinct initial controls encountered in the moving frame of the data path. The time-dependent framework in [1] accommodates this rule. Its single path makes the exact depth accessible.

5 Related work

Pascal parity and fast prediction. The odd-entry count $2^{p(T)}$ is classical. Rowland attributes the binary count to Glaisher and derives it from Lucas's theorem [2]. The fast cellular-automaton prediction work in [1] already uses the Pascal coefficients for homogeneous additive rules. Its modular-sum gates can have unbounded fan-in. The Pascal claim here is a per-time bounded-fan-in lower bound and a matching profile up to a constant factor with a verified uniform De Morgan generator.

Dependent linear tracks and the coefficient formula. Moore's general divide-and-conquer method for time- and space-dependent linear stages gives an $O(\log^2(T+2))$ bounded-fan-in prediction bound in the stated setting [1]. Freezing the controls and using the one-sided stay/hop stencil lets us use prefix statistics and reach $O(\log(T+2))$ in this specialization. The coefficient in Equation 3 also has a classical matrix analogue. The one-step data map is upper bidiagonal on the relevant finite window. Higham's divided-difference formula for functions of such matrices gives the product of intervening off-diagonal entries times a divided difference in the diagonal entries [3]. For the monomial z^T this divided difference is a complete homogeneous polynomial of degree $T - k$. On zero-one diagonal entries it counts the same weak compositions. The polynomial identity can be specialized to integer zero-one entries before reducing modulo two, with no division in $\mathbb{F}_2$.

Mechanization and the meaning of frozen. Silváši and Tomášek formalized bounded grids and computable cellular automata in Lean [4]. Their work supplies precedent for machine-checked cellular automata. The present artifact checks a rule-family depth classification and the actual circuit generator. The freezing model of Ollinger and Theyssier imposes a monotone order on each whole cell state [5]. In Equation 1 only the control word is fixed, and the data at a site can cycle. The comparison identifies antecedents for the main ingredients. Priority for the exact four-way classification and for the transport specialization remains undetermined.

6 Discussion

The frozen classification applies to an explicitly supplied cone and a fixed finite guard pair. Its logarithmic depth is measured against n, which grows linearly with T. Giving a succinct initial state and a binary time query defines a different problem. Uniformity is established for the frozen family. The transport result concerns optimal depth at each window length. The circuit model permits arbitrary wiring. With unbounded fan-in, the transported-control AND can be evaluated in one layer, and the counting argument no longer gives its logarithmic depth. Conversely, if each circuit layer has spatial wiring radius at most r, an essential coordinate at distance T forces $T \le dr$. This additional hypothesis is used for the repository's geometric lower bound in the OR automaton. For unrestricted wiring and fan-in two, that separate example has exact depth $\ell(2T+1)$.

Composition certificate. For an evolution F, a bit-string decoder I into its state space, and an observation O, suppose finite-register Boolean circuits implement initialization, composition along the executed path, and observation. If their depths are $\eta, \delta, \varepsilon$, repeated squaring produces $O\big(F^{T(I(x))}\big)$ from x with depth at most $\eta + 2\ell(T+1)\delta + \varepsilon$. A separate uniform compiler requires logspace generators for the primitives and charges copying during normalization. Its depth bound is $\eta + 1 + 2\ell(T+1)(\delta + 1) + \varepsilon$, with δ bounding both composition primitives. Membership in NC requires a polylogarithmic bound in the actual input length.

Simulation certificate. For a specified system, a logspace transducer from circuit value instances to complete queries $1^T 0x$, together with an observation-equality proof, makes its prediction language P-hard under logspace many-one reductions [6]. A polynomial-time decider for the target yields P-completeness. Assuming $\mathrm{P} \neq$ logspace-uniform NC, a verified hardness witness excludes the target from that parallel class. This conclusion gives no specific numerical depth rate. Rule 110 is a separate concrete P-completeness result in the literature [7].

7 Conclusion

The frozen-control classification separates fixed rules with constant-depth prediction from rules whose depth grows at least along a subsequence. Within the growing regimes, the homogeneous rule's profile follows the set bits of the observation time. The transported model isolates a different mechanism. Distinct initial control sites accumulate along the data path only when the velocities differ. Both classifications follow by identifying the relevant inputs and constructing circuits that combine them with the required depth.

Artifact availability

The Lean development, Typst source, and build instructions are available in the project repository. The Lean toolchain and dependencies are pinned by `lean-toolchain` and `lake-manifest.json`. Run `./scripts/check.sh` at the repository root to check entry-point coverage, build with warnings treated as errors, audit transitive axiom dependencies, and run the audit regression test. Appendix A lists the statements corresponding to the main results.

A Lean correspondence

The paths below are relative to `PredictionDepth/`. The names identify the machine-checked statements underlying the main claims.

Result	Lean module and declaration
Frozen iterate	`FrozenPascal/Semantics.lean` `iterate_data`

Uniform classification	`FrozenPascal/UniformClassification.lean`
	`uniform_depth_classification`
	`predictionLanguage_mem_uniformNC_zero_iff`

Pascal support	`FrozenPascal/HomogeneousPascal.lean` `pascalSupport_card`

Moving controls	`TransportedControl.lean`
	`iterate_data_true_iff`, `essential_prediction_iff`
	`exact_prediction_depth`, `constant_depth_iff`

Composition criterion	`EncodedPathComposition.lean`
	`EncodedPathImplementation.predictor_depth_le`
	`Complexity/UniformComposition.lean`
	`UniformRegisterWitness.depth_le_costs`

Simulation criterion	`Complexity/StandardHardness.lean`
	`SimulationWitness.prediction_P_complete`
	`Complexity/NCClosure.lean`
	`SimulationWitness.prediction_not_uniformNC`

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